

Phase 1 - Updated seed testing protocols for Tomato brown rugose fruit virus and Tomato mottle mosaic virus

We have completed the first phase of our review of Australia's import conditions for tomato and capsicum seed disease testing.

This initial phase focused on evaluating seed testing protocols for Tomato brown rugose fruit virus (ToBRFV) and Tomato mottle mosaic virus (ToMMV), as well as the authorisation process for laboratories conducting these tests.

As a result of this review, we have made 2 key changes to strengthen the seed testing requirements:

- We have established a list of authorised laboratories registered to test tomato and capsicum seed for ToBRFV and ToMMV for export to Australia
- We have updated the list of approved testing protocols for detecting ToBRFV and ToMMV.

These changes aim to improve assurance in laboratory testing and ensure consistent interpretation of results. They will ensure seed is tested using the most suitable protocols and requirements for detecting ToBRFV and ToMMV.

The new requirements became mandatory import conditions on 12 November 2025.

Phase 2 - wider review of seed pathogen testing

We announced the commencement of Phase 2, a wider review of diagnostic testing requirements for tomato and capsicum seed, on 12 May 2026.

This phase builds on Phase 1 and focuses on the remaining regulated seed pathogen tests. The aim is to ensure these tests are technically robust, appropriately validated and fit for purpose to manage biosecurity risk consistent with Australia's Appropriate Level of Protection (ALOP).

Phase 2 of the review is focused on the remaining tomato and capsicum seed pathogens:

- Pepino mosaic virus
- Columnea latent viroid
- Pepper chat fruit viroid
- Potato spindle tuber viroid
- Tomato apical stunt viroid.

The review includes consideration of:

- testing protocols
- interpretation of results
- laboratory authorisation requirements.

Phase 2 does not introduce new seed pathogens and does not include Tomato brown rugose fruit virus (ToBRFV) or Tomato mottle mosaic virus (ToMMV), which were addressed in Phase 1.

Note: There are no immediate changes to the import conditions or testing requirements as a result of Phase 2. Current testing and laboratory authorisation requirements continue to apply until further notice.

Consultation and next steps

We plan to release the draft report for consultation in mid-2027. Please note that this is an indicative timeline and may be subject to change. We will notify permit holders in advance of any proposed changes to import conditions.

Under the updated import conditions from Phase 1, tomato and capsicum seed lots must be tested for ToBRFV and ToMMV by an authorised laboratory.

Laboratories intending to conduct this testing for seed exports to Australia must register to become authorised. To do so, laboratories must complete the registration form below, confirming their agreement to implement the [new testing requirements and protocols](#).

Laboratories must submit their completed form via email to imports@aff.gov.au. Note: please include 'Plant T2 - Seed for sowing – Laboratory authorisation' in the subject line of the email.

The list of authorised laboratories is published on the [Department authorised seed testing laboratories webpage](#).

We have updated the list of approved testing protocols for detecting ToBRFV and ToMMV in tomato and capsicum seed. The new requirements became mandatory import conditions on 12 November 2025.

Some existing seed testing requirements will remain unchanged and continue to apply under the updated protocols. Laboratory test reports for tomato and capsicum seed for export must continue to include the following details:

- The name and address of the testing laboratory
- The full botanical name of the species tested
- The date of testing
- The type of test done
- The seed lot number(s)
- The sample size, which must be 20,000 seeds or 20% of the seed lot for small seed lots
- The subsample size, which must be no greater than 400 seeds per subsample for all testing conducted by PCR
- The pathogens targeted
- The test result confirming freedom from each of the targeted pathogens.

Full testing requirements are available on [BICON](#).

The updated list of approved testing protocols for detecting ToBRFV and ToMMV in tomato and capsicum seed for export to Australia, along with associated requirements, is detailed below:

- Laboratories must use 2 reverse-transcription qPCR protocols to verify the status of ToBRFV and ToMMV in each seed lot.
- For ToBRFV, the CaTa28 protocol must be used as one of the 2 qPCR protocols, and either the CSP1325 protocol or the qs1/p1/qas2 protocol must be used as the second protocol (see [Table 1](#) and [Table 2](#)).
- For ToMMV, the CaTa9 protocol must be used as one of the 2 qPCR protocols, and either the CSP1572 protocol or the ToMMV2 protocol must be used as the second protocol (see [Table 1](#) and [Table 2](#)).
- References to the approved protocols are listed in [Table 3](#). Please note that the department’s requirements for these protocols and for reporting results may differ from some of the parameters stated in the references.
- No other tests are approved for verifying the status of a seed lot regarding these 2 viruses. Laboratories shall not discount a positive result obtained from an approved qPCR test on the basis of any other test result or consideration, except retesting following the ‘[Retesting](#)’ section.

Table 1. Summary of required testing protocols

Virus	First required protocol	Second required protocol
ToBRFV	CaTa28 reverse-transcription qPCR	CSP1325 reverse-transcription qPCR
		OR
		Menzel and Winter reverse-transcription qPCR (qs1/p1/qas2)
ToMMV	CaTa9 reverse-transcription qPCR	CSP1572 reverse-transcription qPCR
		OR
		ToMMV2 reverse-transcription qPCR

[Table 1. Summary of required testing protocols](#)

- The testing protocols that will no longer be approved for use from 12 November 2025 for detecting ToBRFV and ToMMV are listed in [Table 4](#).
- Extraction and amplification positive controls must be run and should include an RNA target. At least one weak positive control should be used that produces a Ct value of 30 or higher.

- At least one negative control should be run. If any negative controls yield Ct values below 35, the testing should be considered invalid, and the department should be contacted.
- A Ct cut-off value of 35 applies to all qPCR tests for ToBRFV and ToMMV. A statement must be included on laboratory reports that states the Ct cut-off point used to determine the result.
- The lowest Ct value obtained for a subsample must be used to interpret the result for that subsample.
- If the Ct value of the subsample is equal to or below 35, this will be considered a positive result, i.e. the target virus has been detected.
- If the Ct value of the subsample is above 35, this will be considered a negative result, i.e. the target virus has not been detected.
- If a laboratory considers that further information is relevant to the interpretation of a set of test results for a seed lot, the laboratory can contact the department for advice. We will consider the information before making a final decision on the status of the seed lot.

A laboratory may re-run a qPCR where the result is considered to be potentially anomalous. For example, if the amplification curve deviates substantially from a typical sigmoidal shape this may be considered anomalous, and the qPCR may be re-run.

- Retesting must be conducted using the seed homogenate (seed slurry) or RNA extract from the same seed subsample.
- An anomalous or ambiguous qPCR result cannot be superseded by testing a different seed sample or subsample.
- At least two additional replicate qPCR reactions must be done for the re-test and the same qPCR protocol that produced the anomalous result must be used for those reactions.
- The results from the retest must be interpreted according to the 'Ct cut-offs and interpretation requirements' specified above.
- Any anomalous finding that has not been shown to be negative by retesting based on the interpretation requirements, must be reported as a positive.
- We may conduct retesting or trace back investigations of seed lots. Laboratories must provide records of previous testing to assist with these investigations.
- Laboratories must keep copies of the quality control records, protocols used and any validation and verification reports relevant to the tests, and must provide these documents upon request by the department .
- Laboratories must keep records of all testing done to meet the Australian requirements for at least 3 years, including Ct values, records of decisions about test results, negative test results, retest results, anomalous results, and results from controls. Images of amplification curves of anomalous results should also be kept.
- If a laboratory resolves an anomalous result and concludes that a seed lot is negative for the virus, the laboratory must record the reason why the anomalous result was considered not to be a detection of the virus.

Table 2.
Approved

**protocols and
result
interpretation
for ToBRFV
and ToMMV,
effective 12
November
2025**

Virus	Protocol	Primers and probe	Date approved
ToBRFV	CaTa28	CaTa28 forward GGTGGTGTCACTGTCTGTTT	24 June 2019
	reverse- transcription qPCR	CaTa28 reverse GCGTCCTTGGTAGTGATGTT CaTa28 6FAM-GAGAATGGAGAGAGCGGACGAGG-BHQ1	
		CSP1325 forward CATTGAAAGTGCATCCGGTTT	
	CSP1325	CSP1325 reverse GTACCACGTGTGTTTGCAGACA	7 June 2019
	reverse- transcription qPCR	CSP1325 VIC*-ATGGTCCTCTGCACCTGCATCTTGAGA- BHQ1 *can be FAM	
ToMMV	qs1/p1/qas2 Menzel and Winter	ToBRFV qs1 forward CAATCAGAGCACATTTGAAAGTGCA ToBRFV qas2 reverse CAGACACAATCTGTTATTTAAGCATC ToBRFV p16	12 November 2025
	reverse- transcription qPCR	6-FAM-ACAATGGTCCTCTGCACCTG- BHQ1	
	CaTa9	CaTa9 forward ATGTGGAGGAACCCTCTATGA	12 November 2025
	reverse- transcription qPCR	CaTa9 reverse AATCTCCTCGCTCCTTGTAAC CaTa9P ROX-TCAATGGCCCGTGGTGAGTTACAA-BHQ1	
	CSP1572	CSP 1572 forward CCCGACTACAGCCGAAACAT	
	reverse- transcription qPCR	CSP 1572 reverse TTAACAGCGGACCTGATCGC	12 November 2025

ToMMV2	ToMMV2 forward	GAAACATTGGATGCCACTCG	
reverse-transcription	ToMMV 2 reverse	CTCTGGTTGTAGAAACCTGTTCC	12 November 2025
qPCR	ToMMV2p FAM-CGATGCTACGGTTGCGATCAGGTC-BHQ1		

Table 2. Approved protocols and result interpretation for ToBRFV and ToMMV, effective 12 November 2025

Table 3. References to protocols

Protocol	Reference
CaTa28	
CSP1325	
Menzel and Winter	
CaTa9	
CSP1572	
ToMMV2	

Table 3. References to protocols

Table 4. PCR protocols removed from the approved list

Protocol and Primers	Reason	Approval period
Levitzky et al. 2019		
F GAAGAAGTTGTTGATGAGTTCAT R GATTTAAGTGGAGGGAAAAACAC Reference Levitzky, N, Smith, E, Lachman, O, Luria, N, Mizrahi, Y, Bakelman, H, Sela, N, Laskar, O, Milrot, E & Dombrovsky, A, 2019, 'The bumblebee <i>Bombus terrestris</i> carries a primary inoculum of Tomato brown rugose fruit virus contributing to disease spread in tomatoes' PloS one, volume14, issue 1, p.e0210871.	The protocol of Levitzky et al. 2019 is not suitable for the detection of ToBRFV and ToMMV as it is not sufficiently sensitive and uses a group specific primer set.	1 April 2019 - 12 November 2025
Alkowni et al. 2019		
F AATGTCCATGTTTGTACGCC R CGAATGTGATTAAACTGTGAAT	The protocol of Alkowni et al. 2019 is not suitable for the detection of ToBRFV as it is not sufficiently sensitive.	1 April 2019 - 12 November 2025

Reference

Alkowni, A, Alabdallah, O & Fadda, Z 2019, 'Molecular identification of tomato brown rugose fruit virus in tomato in Palestine' Journal of Plant Pathology, available at DOI 10.1007/s42161-019-00240-7.

Li et al 2018

TobamodF TKGAYGGNGTBCCNGGNTGYGG
TobamodR ACNGAVTBNABCTGTAATTGCTAT

Reference:

Li, Y, Tan, G, Lan, P, Zhang, A, Liu, Y, Li, R, & Li, F, 2018, 'Detection of tobamoviruses by RT-PCR using a novel pair of degenerate primers' Journal of virological methods, volume 259, pp.122-128.

Test results indicate that the protocol of Li et al. (2018) is not sufficiently sensitive. The protocol was removed from the approved list on 7 June 2019.

1 April 2019
- 7 June
2019

Table 4. PCR protocols removed from the approved list

Source: <https://www.agriculture.gov.au/biosecurity-trade/import/goods/plant-products/seeds-for-sowing/pathogen-testing-req-tomato-capsicum>